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SLOT - D

SIMATS ENGINEERING
ASSESSMENT TOOL 2

COURSE NAME: CSA07 **COURSE CODE:** Computer Networks

COURSE OUTCOME COVERED

CO1: Apply the functions of OSI layer in enabling reliable data transmission across networks. (BL2, BL3)

Assessment Tool Weightage: 30%

Annexure A

CONCEPT MAPPING

Code of Conduct:

I S. Nirmal (19251249) *(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: S. Nirmal

Criteria	Weightage (%)	Q1 10marks	Q2 10marks	Q3 10marks
Concept Identification	20%	2	2	2
Concept Mapping Structure	25%	2	1	2
Linking & Relationships	20%	1	2	1
Application & Analysis	20%	1	1	1
Explanation & Clarity	15%	1	1	1

Total Marks 21 /30

Signature of the Faculty

CONCEPT MAPPING QUESTIONS

1. A partially completed concept map is given below:

Application → Presentation → _____ → Transport → Network → _____ → Physical.
Analyze the missing layers and explain how their functions are essential for reliable communication.

2. Construct a concept map linking data units (Segment, Packet, Frame, Bits) with their respective OSI layers and analyze how transformation of data units supports transmission.

3.Given the concept map:

DataLink → MACAddress → LocalDelivery

Network → IP Address → Global Delivery

Explain how these relationships ensure complete data delivery across networks.

4. A concept map shows:

ErrorControl

→ DataLinkLayer → ErrorDetection

→ Transport Layer → Error Correction

Analyze how the interaction between these layers improves reliability in data transmission.

5. Develop a concept map for web browsing using OSI layers and analyze how failure in one layer affects the entire communication process.

6. A network issue is represented in the concept map:

Physical ✔ → Data Link ✔ → Network ✖ → Transport ✖ → Application ✖

Analyze the problem and explain how the concept map helps in troubleshooting.

RUBRICS FOR CALCULATION

Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Concept Identification	20%	Accurately identifies all relevant OSI layers, concepts, and relationships	Identifies most concepts with minor omissions	Identifies limited concepts; some inaccuracies	Unable to identify key concepts correctly
Concept Mapping Structure	25%	Constructs a clear, logical, and well-organized concept map with proper hierarchy and connections	Concept map is mostly clear with minor structural issues	Concept map lacks clarity, organization, or proper flow	No clear structure or incorrect mapping
Linking & Relationships	20%	Clearly establishes correct relationships between layers, functions, and processes	Relationships are mostly correct with minor errors	Limited or partially correct relationships	Incorrect or missing relationships
Application & Analysis	20%	Effectively applies concepts to explain processes (e.g., encapsulation, error control) with strong reasoning	The application is correct but lacks depth in analysis	Basic application with weak explanation	Unable to apply concepts correctly
Explanation & Clarity	15%	Provides a clear, concise, and well-structured explanation supporting the concept map	Explanation is understandable with minor gaps	Explanation is unclear or incomplete	No clear or meaningful explanation

CSA0710 - Computer Networks

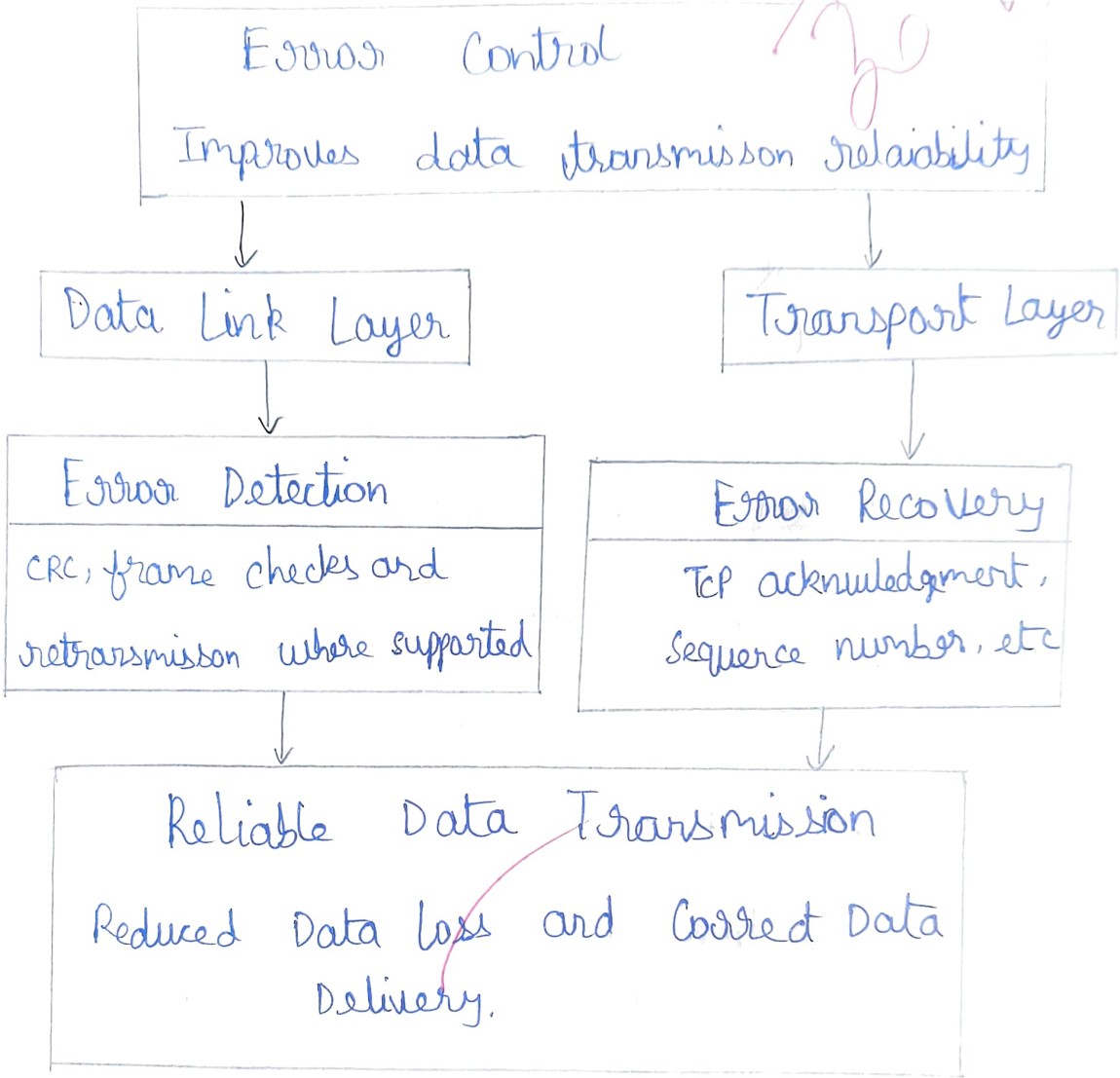
CO1 AT-2 Concept mapping (27)

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Concept Map:



Analysis:

⇒ The Data link layer detects crouppted frames using techniques such as CRC. Depending on the protocol, damaged frames may be retransmitted.

→ The transport layer, particularly TCP, uses sequence numbers, acknowledgements and retransmissons to recover missing data.

⇒ The data link layer handles errors on individual links, while the transport layer provides end-to-end reliability.

⇒ Conclusion: both layers work together to reduce data loss and ensure reliable communication.

(5) Concept map:

7. Application Layer
HTTP/HTTPS
↓
6. Presentation Layer
Data formatting/Encryption
↓
5. Session Layer
Session Managment
↓
4. Transport Layer
TCP/UDP & ports
↓
3. Network Layer
IP and routing
↓
2. Data link Layer
Frames & MAC
↓
1. Physical Layer
Transmisson of bits

Analysis:

=) All seven OSI layer cooperate during Web browsing. The Application layer handles web requests, the transport layer delivers data, the Network layer routes packages and the lower layers transmit them in the network.

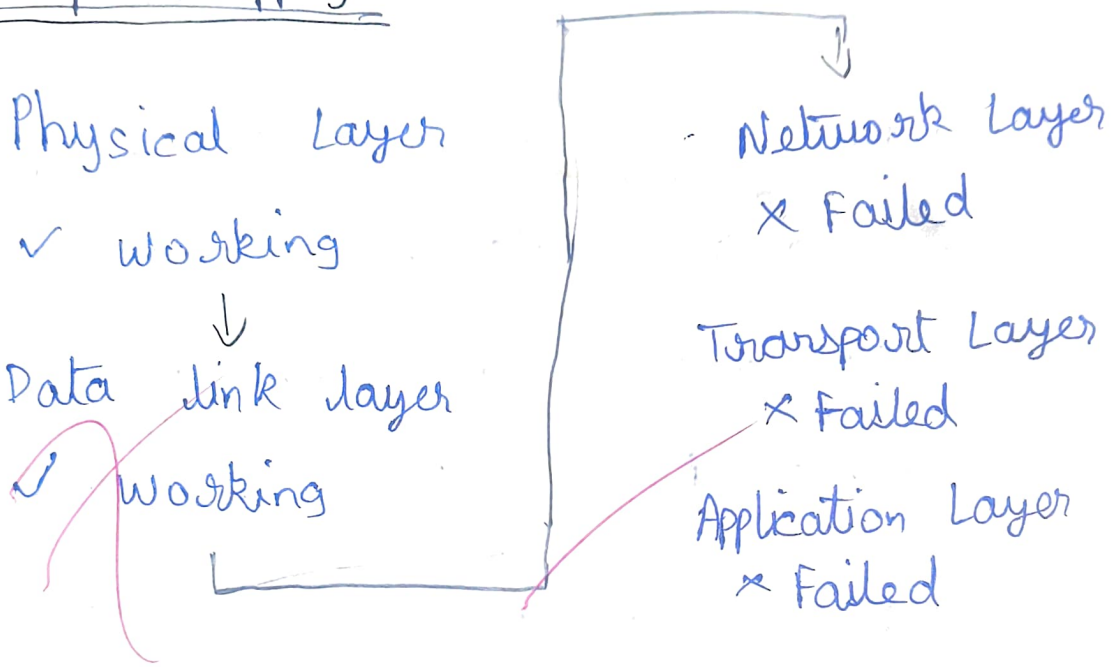
=) Failure in any essential layer can interrupt communication.

=) Conclusion:

The OSI Layers work together to enable web browsing. A failure in one layer can affect the entire communication process.

(6) Network trouble shooting.

Concept mapping:



Analysis:

→ The physical and data link layer are working.

→ Network layer is the frist identified failure. Possible causes are incorrect IP or missing gateway.

→ Transport and Application Layers also failed.

Conclusion: Fixing the root cause can restore communication, provided no other fault exist.